

HemoScan AI - Clinical Hematology Report

Hematological Evaluation Report

Patient Information

- * Name: Piyush Kumar Singh
- * Age: 21.0 Years
- * Gender: Male
- * Phone: 9876543210

Clinical Data Comparison

The following table presents the patient's Complete Blood Count (CBC) indices compared against standard clinical reference ranges for an adult male.

Parameter	Patient Value	Reference Range (Adult Male)	Status
Hemoglobin (Hb)	11.5 g/dL	13.5 – 17.5 g/dL	Low
RBC Count	4.1 million/ μ L	4.5 – 5.9 million/ μ L	Low
PCV (Hematocrit)	34.2 %	41.0 – 50.0 %	Low
MCV	82.0 fL	80.0 – 100.0 fL	Normal (Borderline)
MCH	26.5 pg	27.0 – 33.0 pg	Low
MCHC	31.5 g/dL	32.0 – 36.0 g/dL	Low

Clinical Interpretation

The laboratory findings for Mr. Piyush Kumar Singh indicate a clear reduction in the total red blood cell mass and hemoglobin concentration. The Hemoglobin level of 11.5 g/dL falls significantly below the lower threshold for adult males (13.5 g/dL), confirming a state of anemia. This is further supported by the PCV (34.2%) and RBC count (4.1 million/ μ L), both of which are below optimal physiological levels, suggesting a decreased oxygen-carrying capacity of the blood.

A detailed analysis of the erythrocyte indices reveals a Mean Corpuscular Volume (MCV) of 82.0 fL, which is at the lower boundary of the normal range. However, the MCH (26.5 pg) and MCHC (31.5 g/dL) are both slightly depressed. This combination suggests that while the red blood cells are of relatively normal size (normocytic), they contain a lower-than-optimal concentration of hemoglobin (hypochromic). This "normocytic-hypochromic" or "borderline microcytic" pattern is frequently observed in the early stages of nutritional deficiencies or as a secondary response to underlying systemic factors.

Risk Level

Moderate

The patient is currently symptomatic of mild-to-moderate anemia. While not in an acute critical range, an Hb of 11.5 g/dL in a 21-year-old male requires clinical intervention to prevent further decline, which could lead to significant fatigue, reduced exercise tolerance, and cognitive impairment.

Possible Anemia Type

Based on the indices provided, the differential diagnosis includes:

1. Early-Stage Iron Deficiency Anemia (IDA): The most common cause, where MCH/MCHC drop before the MCV significantly decreases.

2. Anemia of Chronic Disease (ACD): Often presents as normocytic/normochromic but can trend toward hypochromic.

3. Thalassemia Trait: Though MCV is usually much lower in Thalassemia, it remains a secondary consideration in specific demographics.

Suggested Diagnostic Tests

To pinpoint the etiology of the anemia and initiate appropriate therapy, the following investigations are recommended:

* Serum Iron Profile: Including Serum Iron, Total Iron Binding Capacity (TIBC), and Serum Ferritin levels to assess iron stores.

* Peripheral Blood Smear (PBS): To evaluate RBC morphology (checking for anisocytosis or poikilocytosis).

* Reticulocyte Count: To determine the bone marrow's regenerative response.

* Vitamin B12 and Folate Levels: To rule out concurrent nutritional deficiencies.

* Stool Routine/Occult Blood: To rule out any gastrointestinal blood loss, which is a common cause of iron depletion in young adults.